Assignment4

Weikang Kong
ID 217655

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1 Unet reconstruction and K-means clustering

1.1 Train model directly

The dataset was split into 80% training data and 20% testing data, and 10% of the training set was further used as a validation set for early stopping.

U-Net performs well as an autoencoder for the reconstruction task (MSE $\approx 1.8 \times 10^{-4}$), and the reconstructed waveforms are highly similar to the original seismic waveforms (Figure 1). This indicates that the model has successfully learned a compact low-dimensional representation of the data. After applying PCA followed by K-means in the bottleneck feature space, one event type (label 0) is highly concentrated in the latent space and forms a very pure cluster. In contrast, the other event type (label 1) shows substantial internal structural variation in the latent space and is split across two clusters, which leads to an ARI of about 0.21, only slightly better than random assignment. A likely reason is that the autoencoder objective emphasizes overall reconstruction accuracy but does not explicitly enforce class separability. As a result, the bottleneck features encode both class-related information and non-class factors (e.g., amplitude variations and noise patterns). K-means therefore tends to partition along directions of maximum global variance rather than along the true class boundary.

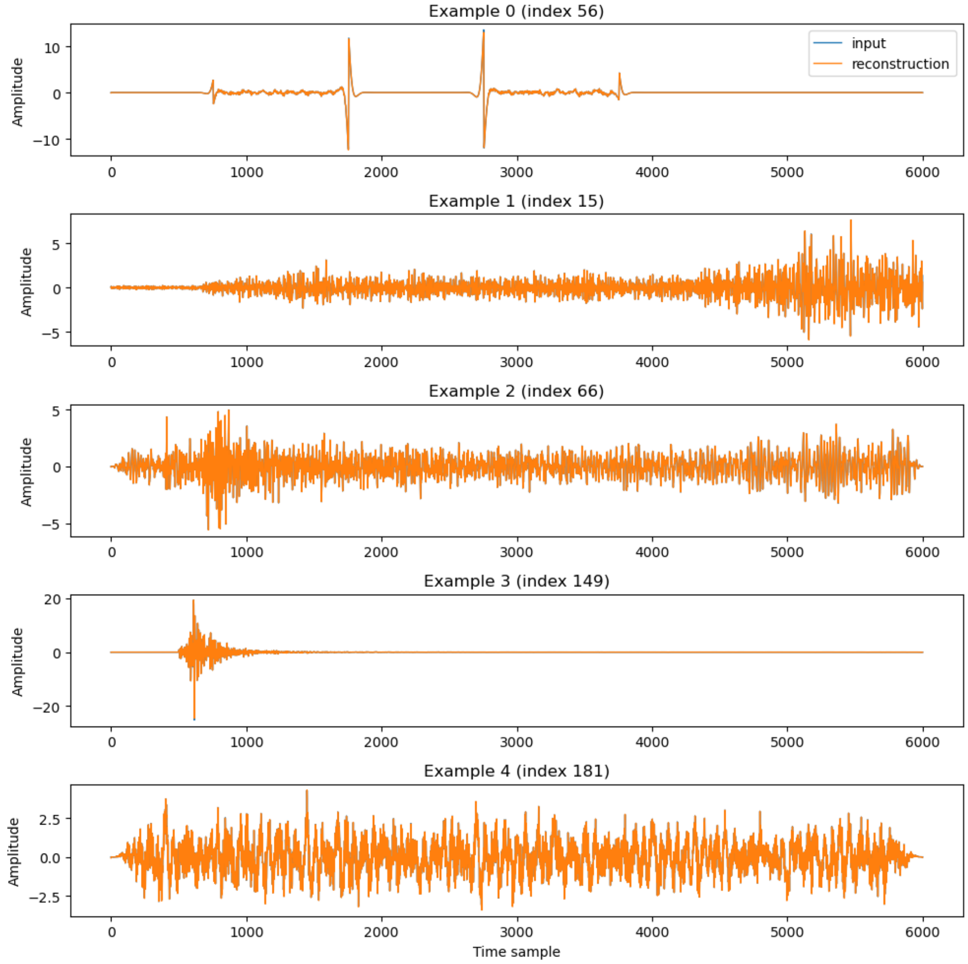

Figure 1: Reconstruction result from U-Net model.

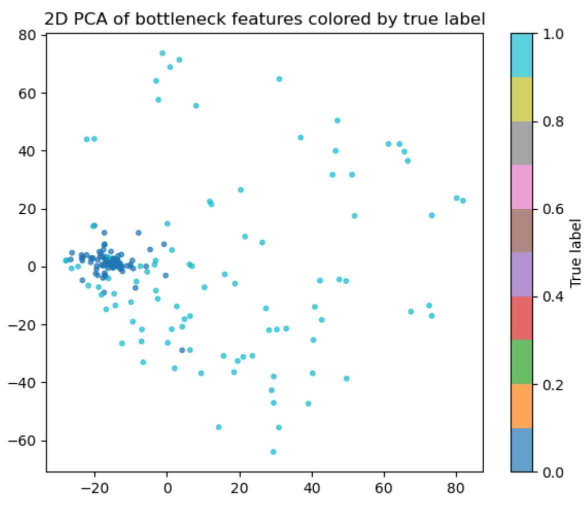

Figure 2: Real-label classification results.

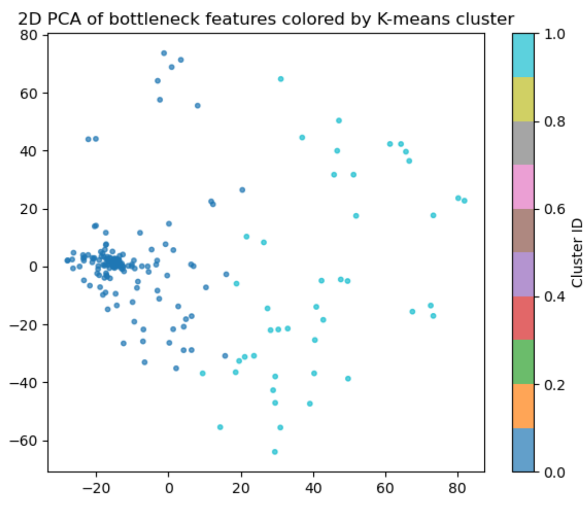

Figure 3: K-means clustering results.

1.2 Frequency transformed

In this experiment, Fast Fourier Transform (FFT) was applied. The magnitude spectrum was then obtained by taking the absolute value of the complex Fourier coefficients, and used as the input representation for subsequent model training. The latent features were reduced to 10 dimensions using Principal Component Analysis (PCA) before applying K-means clustering.

Using frequency-inputs led to worse clustering performance (ARI, $\approx -1.9 \times 10^{-5}$) (Figure 5 & Figure 6), likely because magnitude spectra (e.g., the absolute value of the FFT) introduce a larger dynamic range and a more skewed, heavy-tailed distribution than the already normalized time-domain waveforms. With a MSE objective, training can be dominated by a few high-energy low-frequency or direct current components, so the model under-learns discriminative bands relevant to event types. In addition, magnitude-only features discard phase information, while phase and temporal structure often encode key cues such as arrival timing and waveform shape. As a result, the learned latent representations are less separable, leading to poorer clustering performance.

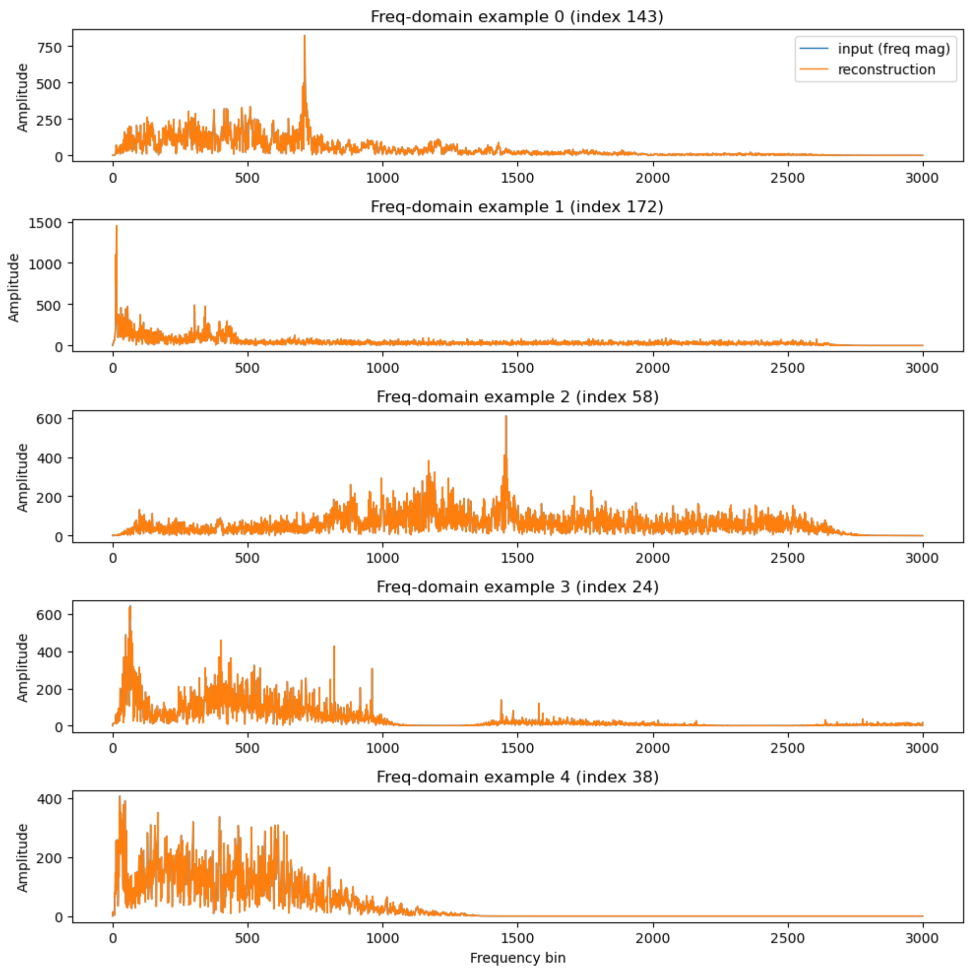

Figure 4: Reconstruction result after frequency transformed from U-Net model.

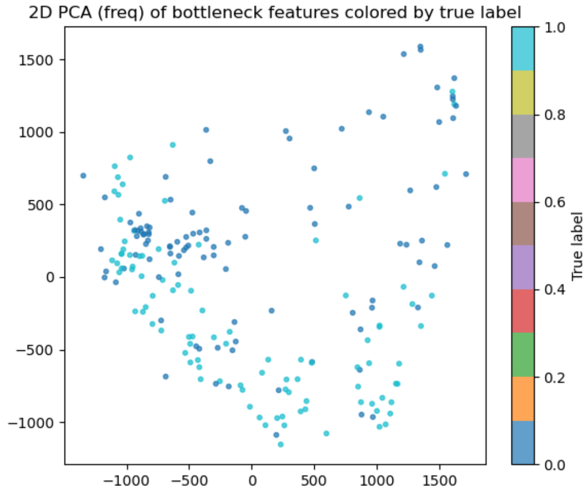

Figure 5: Real-label classification results after frequency transformation.

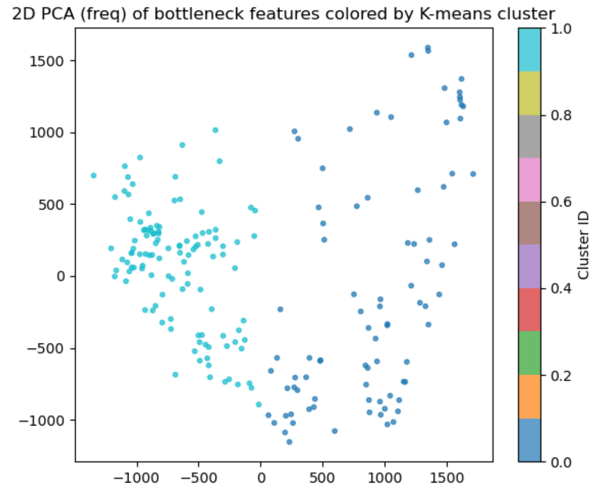

Figure 6: K-means clustering results after frequency transformation.

1.3 Denoising and Interpolating Marmousi2 Data

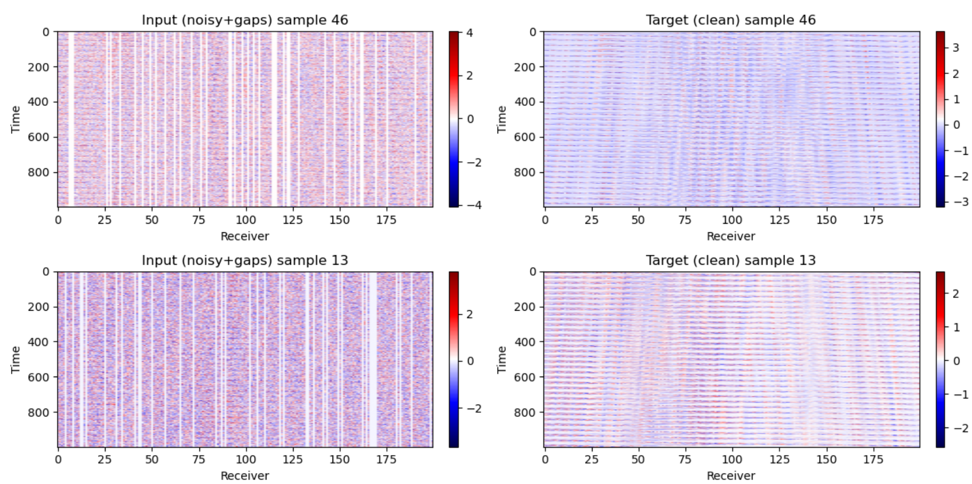

Figure 7: Example of noising and clean results on Marmousi2 data.

In the Marmousi2 denoising experiment, the input per-sample signal-to-noise ratio (SNR) values were −3.57 dB, −8.80 dB, and −5.13 dB, while the output SNR values increased to 3.60 dB, 1.22 dB, and 2.86 dB. This corresponds to SNR improvements of 7.16 dB, 10.03 dB, and 7.99 dB for the three samples, respectively, with a mean improvement of 8.39 dB, indicating substantial denoising effectiveness (Figure 8).

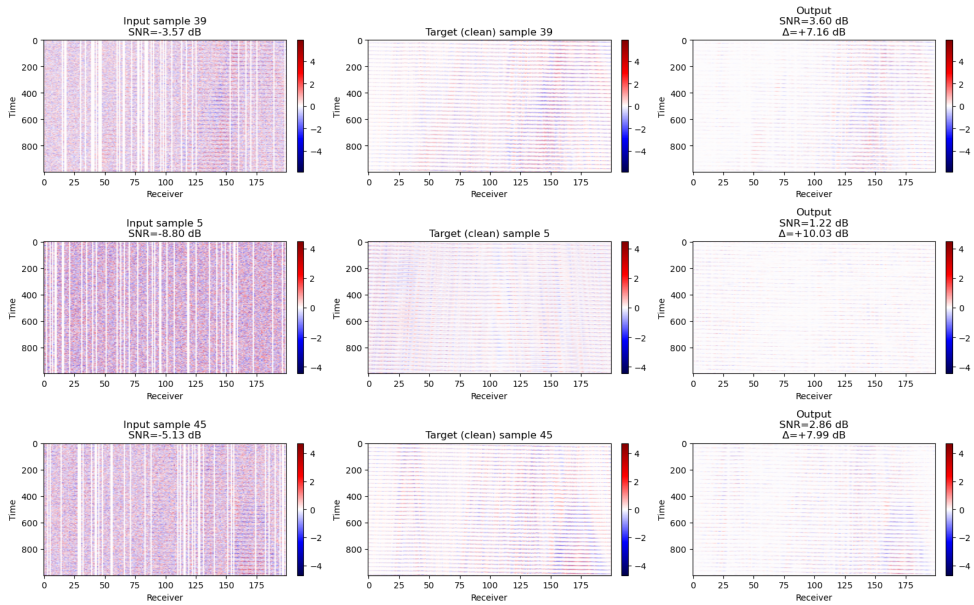

Figure 8: Some examples of denoising performance comparison and SNR improvement on Marmousi2 data.

2 Code

https://github.com/WKKKKKKKKKKKKKK/ML_IN_GEO